CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The HCF of 24, 36 and 60 is: 1

(a) 11 **(b)** 12 **(c)** 14 **(d)** 13

2. If α, β are zeroes of $x^2 - 7x + 10$, then $\alpha^2 + \beta^2$ is: 1

(a) 29 **(b)** 28 **(c)** 30 **(d)** 49

3. The value of x in $7x + 2y = -6$, $3x + 5y = 14$ is: 1

(a) -1 **(b)** -2 **(c)** 0 **(d)** -3

4. The sum of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) $\frac{9}{2}$ **(b)** $\frac{11}{2}$ **(c)** $\frac{13}{2}$ **(d)** $\frac{7}{2}$

5. The LCM of 6, 8 and 12 is: 1

(a) 25 **(b)** 24 **(c)** 23 **(d)** 57

6. If α, β are zeroes of $x^2 - 5x + 6$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is: 1

(a) $-\frac{5}{6}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{6}{5}$ **(d)** $\frac{11}{6}$

7. The value of y in $11x + 4y = 53$, $5x + 3y = 30$ is: 1

(a) 7 **(b)** 4 **(c)** 6 **(d)** 5

8. The product of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) 2 **(b)** 3 **(c)** 1 **(d)** 4

9. The largest number that divides 70 and 125 leaving remainders 5 and 8 respectively is: 1

(a) 14 **(b)** 13 **(c)** 15 **(d)** 12

10. If $x = 2$ is a zero of $x^2 - 7x + k$, then k is: 1

(a) 9 **(b)** 12 **(c)** 10 **(d)** 11

11. The pair $2x + ky = 6$, $4x + 4y = 12$ has infinitely many solutions when $k =$ 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

12. The roots of $x^2 - 11x + 18 = 0$ differ by: 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 9

13. The least number which when divided by 6, 8 and 12 leaves remainder 3 each time is: 1

(a) 27 **(b)** 24 **(c)** 21 **(d)** 28

14. The value of k for which $x = 2$ is a zero of $x^2 + kx - 10$ is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

15. For the pair $kx + 2y = 5$ and $6x + 1y = 3$ to have a unique solution, k must NOT equal: 1

(a) 12 **(b)** 14 **(c)** 13 **(d)** 11

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the largest number which divides 245 and 1029 leaving remainders 5 and 5 respectively. 2

17. If α, β are zeroes of $x^2 - 8x + 12$, find $\alpha^2 + \beta^2$. 2

18. Solve by the cross-multiplication method: $7x + 2y = -6$ and $3x + 5y = 14$. 2

19. Solve for x: $2x^2 - 9x + 4 = 0$ by the quadratic formula. 2

20. Find the smallest number which leaves remainder 5 when divided by 12, 16 and 18. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. The HCF of two numbers is 9 and their LCM is 360. If one number is 72, find the other. 3

22. If α, β are zeroes of $x^2 - 6x + 8$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ and $\alpha^2 + \beta^2$. 3

23. Solve by elimination: $7x + 2y = -6$ and $3x + 5y = 14$. 3

24. Solve for x: $\frac{1}{x} + \frac{1}{x+1} = \frac{5}{6}$. 3

OR

Three bells toll at intervals of 12, 15 and 18 minutes. If they toll together at 8:00 a.m., after how many minutes will they next toll together? 3

25. Find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of $x^2 - 5x + 6$. 3

OR

The sum of the digits of a two-digit number is 11. If the digits are interchanged, the number becomes 45 more. Find the number. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two tankers contain 850 litres and 680 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

OR

Find all the zeroes of $x^3 - 6x^2 + 11x - 6$ if two of its zeroes are 1 and 2. 5

27. A father is 10 times as old as his son. After 6 years the father will be 4 times as old as the son will be. Find their present ages. 5

OR

A train travels 200 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 2 hours less for the journey. Find the original speed. 5

END OF QUESTION PAPER — **Total: 50 Marks**